TEXORA · LECTURE SHEET

Grey Fabric Calculations

Before a fabric is woven it exists as a specification: ends and picks per inch, two yarn counts, two crimps. This sheet turns that specification into the numbers the mill trades on — warp and weft weight per square metre, GSM, cover factor, and weight per linear metre — and shows how a buyer can check a delivered fabric against its own spec.

FM-3207 · ADVANCED WEAVING AND KNITTING CALCULATIONS · TOPIC 5 · SPECIALIZATION — FABRIC MANUFACTURING

1. The master formula

One square metre of fabric contains EPI × 39.37 warp ends (each longer than a metre by its crimp) and PPI × 39.37 weft picks (each longer than a metre by its crimp). Multiply length by tex and the weights fall out:

$$\text{Warp weight (g/m}^2\text{)} = \text{EPI} \times 39.37 \times (1 + \text{warp crimp}) \times \text{warp tex} \div 1{,}000$$

$$\text{Weft weight (g/m}^2\text{)} = \text{PPI} \times 39.37 \times (1 + \text{weft crimp}) \times \text{weft tex} \div 1{,}000$$

$$\text{GSM} = \text{warp weight} + \text{weft weight}$$

WORKED EXAMPLE — THE COURSE REFERENCE FABRIC

Plain weave: EPI 100, PPI 80, warp and weft both Ne 30 (19.68 tex), warp crimp 8%, weft crimp 5%.

Warp = 100 × 39.37 × 1.08 × 19.68 ÷ 1,000 = **83.7 g/m²**.

Weft = 80 × 39.37 × 1.05 × 19.68 ÷ 1,000 = **65.1 g/m²**.

GSM = 83.7 + 65.1 = 149 g/m².

At a grey width of 1.5 m the fabric weighs 149 × 1.5 = **223 g per linear metre** — warp 125.6 g and weft 97.6 g of it (the split Sheet 13 costs, item by item). This fabric is the reference cloth for the whole FM-3207 costing and planning chain.

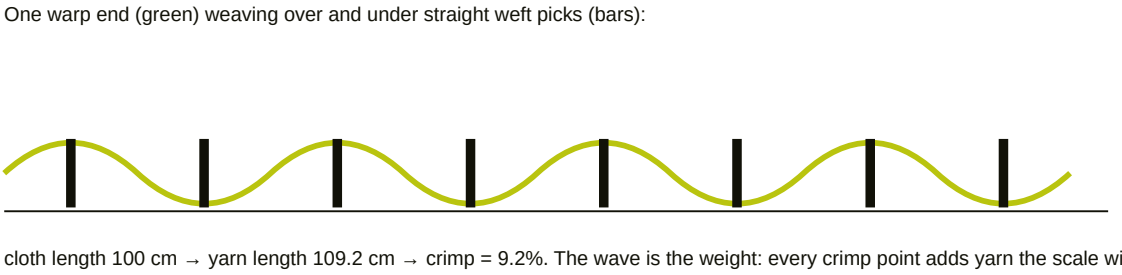


Figure 1 — Crimp: the warp's wavy path through straight weft is why yarn length always exceeds cloth length.

2. Crimp — measured, not assumed

Crimp is measured by extracting a marked length of yarn, straightening it under standard tension and comparing lengths: a warp end measuring 109.2 cm from 100 cm of cloth carries (109.2 − 100)/100 = **9.2% crimp**. Warp crimp normally exceeds weft crimp in loom-state fabric (the warp does most of the bending around a comparatively straight weft), and the two swap partly at finishing as tensions relax. Every GSM dispute is, at bottom, a crimp dispute: 1% of uncounted warp crimp on the reference fabric is 0.78 g/m² — invisible in the cloth, visible in a 10,000 m invoice.

3. Cover factor

Weight is not cover: a fabric can be heavy and open, or light and closed. Peirce's cover factor compares thread spacing with yarn diameter (diameter ∝ 1/√Ne):

$$\text{Cover factor} = \text{threads per inch} \div \sqrt{Ne}$$

WORKED EXAMPLE — COVER OF THE REFERENCE FABRIC

Warp CF = 100 ÷ √30 = **18.3**; weft CF = 80 ÷ √30 = **14.6**.

Cloth cover factor ≈ $CF_1 + CF_2 - (CF_1 \times CF_2)/28$ = 18.3 + 14.6 − 9.5 = **23.3** (on the conventional scale where 28 is theoretical full cover).

The warp dominates this cloth's cover — correct for a warp-faced working fabric — and the gap between 18.3 and 14.6 tells the designer which direction to change if the fabric must close up or open out.

4. Checking a delivered fabric

The same formulas audit production: measure EPI/PPI on the cloth, unravel and weigh warp and weft from a known area, extract crimp, and rebuild the GSM from the measurements. A delivered GSM 4% under specification with correct counts and densities is a crimp story (loom settings, take-up); with

correct crimp it is a count story (yarn substitution); with low PPI it is a picking story. The calculation does not just describe the fabric — it assigns the fault.

Glossary

Grey (greige) fabric — fabric as it leaves the loom, unfinished. **Crimp** — extra yarn length in the cloth from interlacing, % over the cloth length. **Cover factor** — spacing-to-diameter ratio of a thread system (Peirce). **GSM** — fabric mass per unit area, g/m². **Linear metre weight** — GSM × width in metres.

Key points

• GSM = EPI × 39.37 × (1+crimp) × tex ÷ 1,000, warp plus weft (worked: 83.7 + 65.1 = 149 g/m²; 223 g per linear metre at 1.5 m).
• Crimp is measured from extracted yarn (109.2 cm in 100 cm → 9.2%) and is where GSM disputes live.
• Cover factor = threads/inch ÷ √Ne (worked: warp 18.3, weft 14.6, cloth ≈ 23.3).
• The formulas run backwards as an audit: GSM error decomposes into density, count or crimp.

Practice

1. Compute the GSM of a plain weave with EPI 120, PPI 90, warp and weft Ne 40 (14.76 tex), warp crimp 10%, weft crimp 6%.
2. The reference fabric is rewoven with weft Ne 26 (22.71 tex) at unchanged densities and crimps. Find the new GSM.
3. A fabric specified at 149 GSM is delivered at 143 GSM with EPI, PPI and counts all correct. Which single measurement do you take first, and what result would explain the shortage?

Answer key

1. Warp = 120 × 39.37 × 1.10 × 14.76 ÷ 1,000 = 76.7; weft = 90 × 39.37 × 1.06 × 14.76 ÷ 1,000 = 55.4; GSM = **132 g/m²**.
2. Weft = 80 × 39.37 × 1.05 × 22.71 ÷ 1,000 = 75.1; GSM = 83.7 + 75.1 = **159 g/m²**.
3. Measure crimp (warp first — it carries the larger weight share). A warp crimp of about 3% instead of 8% removes 100 × 39.37 × 0.05 × 19.68 ÷ 1,000 ≈ 3.9 g/m², and a similar small weft shortfall closes the gap — the fabric was woven flatter than specified, typically a take-up/tension setting, not a yarn fault.